KENYATTA UNIVERSITY

EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE

(CIVIL ENGINEERING)

ECV 209: THEORY OF STRUCTURES

1ST SEMESTER 2021/2022 **TIME: 2 HOURS**

INSTRUCTIONS

(a) This paper contains **FIVE (5)** questions.

(b) You are required to answer **THREE (3)** questions only.

Q.1 Fig Q.1 Shows a cable .Determine: [30 Marks]

(a) Tension in each cable segment

(b) Distance y_D.

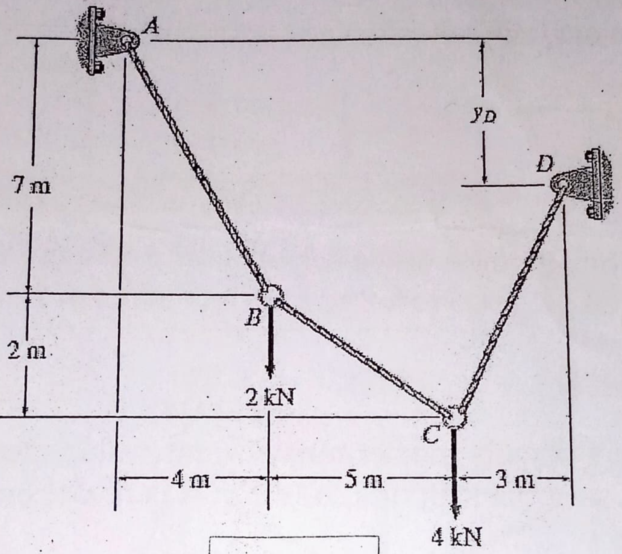

Fig Q.1

Q.2 The beams *AB* and *BC* are supported by the cable that has a parabolic shape. Determine the tension in the cable at points *D*, *F*, and *E*, and the force in each of the equally spaced hangers. [30 Marks]

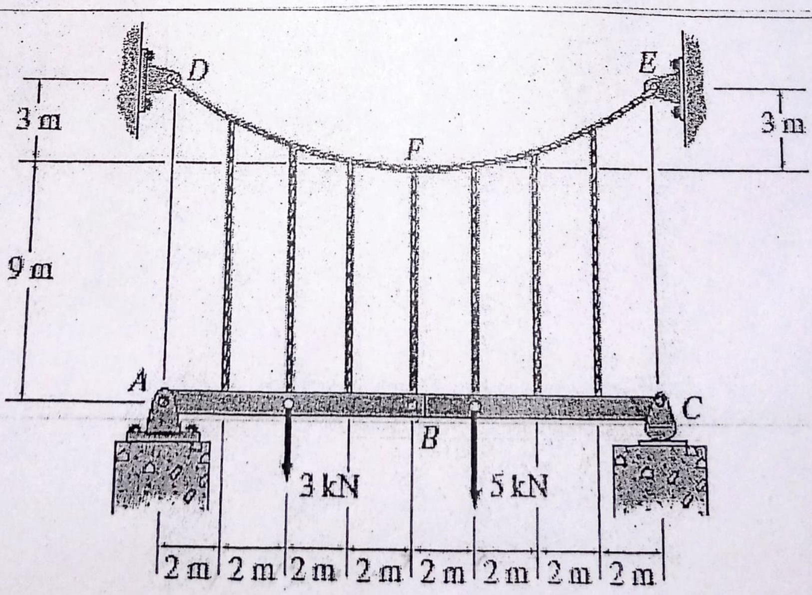

Fig Q2

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

Q.3 Fig Q.3 shows a determinate beam fixed at one end. Assume A is fixed. [30 Marks]
Draw the influence line for

(a) the vertical reaction at A,
(b) the shear at B, and
(c) the moment at B.

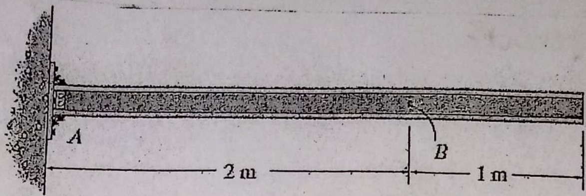

Fig. Q3

Q.4 Fig Q.5 4 ,Shows a simply supported beam. EI is constant. [30 Marks]

(a) Determine the equations of the elastic curve for the beam using the and coordinates. double integration method } specify the co-ordinates
(b) Specify the slope at A and
(c) the maximum deflection.

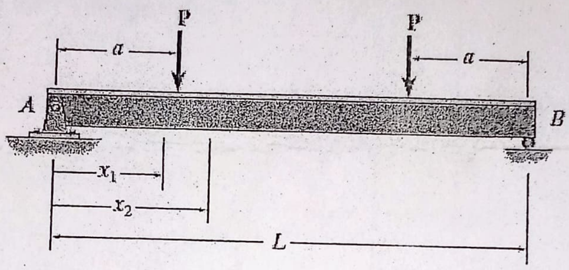

Fig.Q4

Q.5. Fig Q. 5 shows a statically indeterminate frame, using method of consistent deformations, Determine the reactions at the supports. Assume A is fixed connected and E is constant.

[30 Marks]

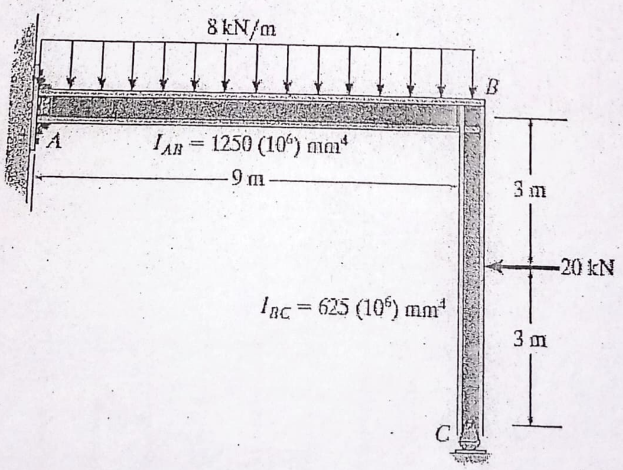

Fig Q.5

KENYATTA UNIVERSITY

UNIVERSITY EXAMINATIONS 2013/2014

SECOND SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF SCIENCE IN CIVIL ENGINEERING

ECV 209: THEORY OF STRUCTURES II

DATE: Wednesday, 16th April 2014 **TIME: 11.00 a.m. – 1.00 p.m.**

INSTRUCTIONS

Question **ONE** is compulsory (30 Marks)
Answer any other **TWO** questions (20 marks)

SECTION A (30 MARKS) COMPULSORY

1. (a) i. Define an arch (1 mark)

ii Explain the difference between statically determinate and indeterminate structures. (3 marks)

iii Give reasons why statically indeterminate structures are preferred to determinate structures. (6 marks)

(b) The three-hinged arch is subjected to the loading shown in Fig. Q1. Determine the forces in members CH and CB. GF is intended to carry no force. (20 marks)

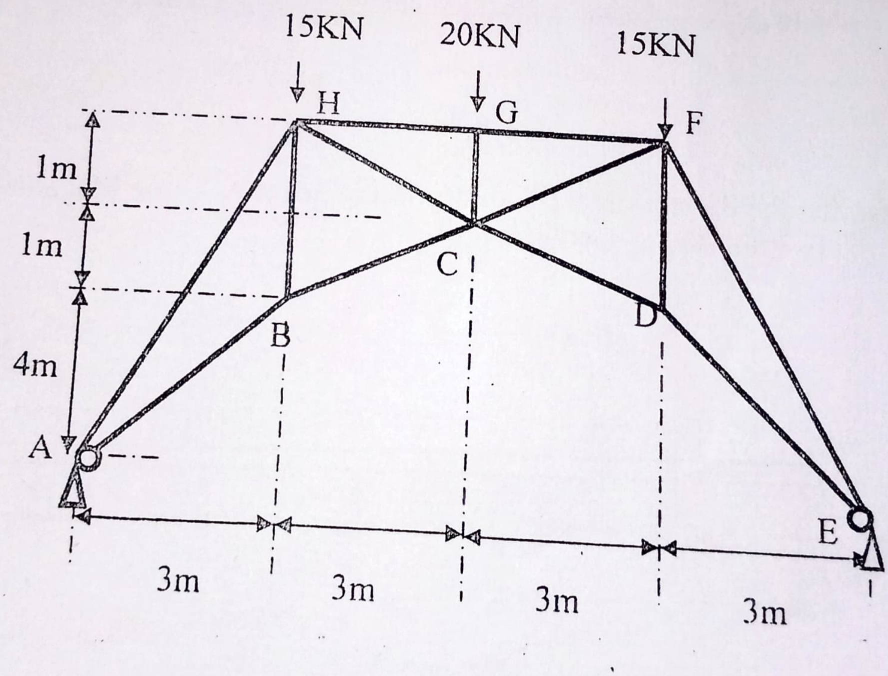

Fig. Q 1 (Drawn not to scale)

INVOLVEMENT IN ANY EXAMINATION IRREGULARITY SHALL LEAD TO DISCONTINUATION

SECTION B(20 Marks each) Answer any **two** questions

2. Determine the maximum positive shear created at point B in the beam bridge shown in Fig. Q 2 due to the wheel loads of the moving truck. (20 marks)

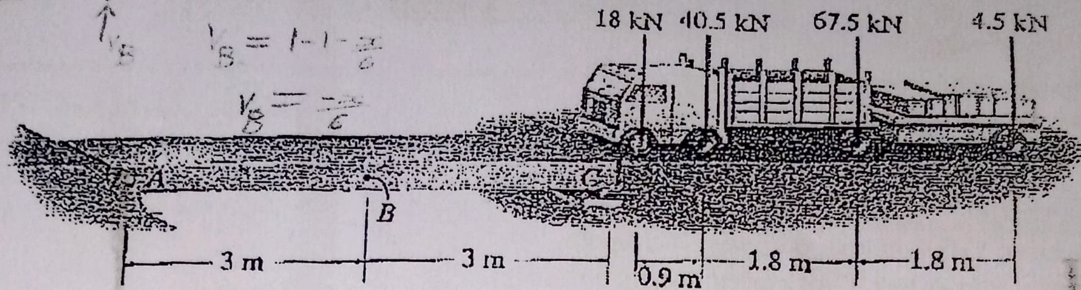

Fig. Q2

3. Determine the reactions for the continuous beam shown in Fig. Q 3 due to the uniformly distributed load and support settlement of 10mm at A,50mm at B, 20mm at C and 40mm at D. Use the three- moment equation. E=200Gpa and I=700(10^6) mm^4 (20 marks)

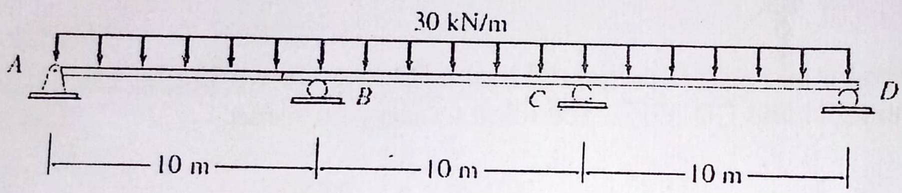

Fig. Q3

4. Determine the deflection at the mid-point of beam AB. Use Macaulay method.
E=208GN/m^2. I=82x10-6m^4 (20 marks)

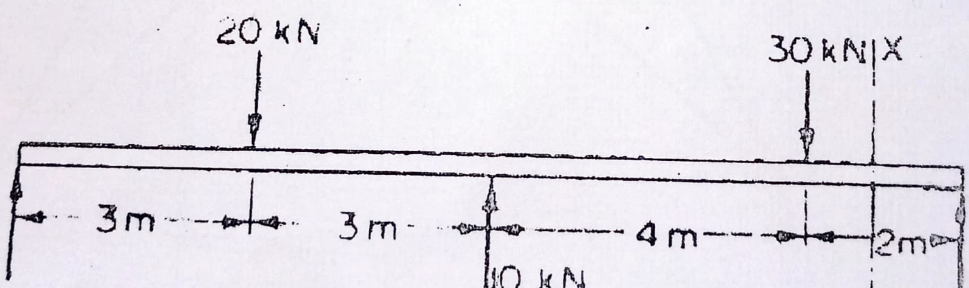

Fig. Q4

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